

RODRIGO GOMES DE ARAÚJO

☎ +351 914 574 743 — ✉ contact@rodrigoaraujo.pt — 🌐 rodrigoaraujo.pt

EDUCATION

Master's in Computer Science

Faculdade de Ciências da Universidade do Porto (FCUP)

Sep 2025 – Present (Expected 2027)

Bachelor's in Computer Science and Engineering

Faculdade de Engenharia da Universidade do Porto (FEUP)

Sep 2022 – Jul 2025

PROJECTS

PoW blockchain for decentralized auctions — Rust, Tokio, Dalek, libp2p, Tonic, Docker

- Developed a DHT overlay based on Kademlia for **decentralized network construction** and peer discovery.
- PoW Blockchain with **Blake2b** hashing, **Ed25519** signatures, and **Merkle Trees** for transaction validation.
- Block propagation through GossipSub. **Chain catch-up** and **block recovery** through request-response. Designed and implemented a deterministic **longest chain strategy** for fork resolution.
- **Client for auctions** using the blockchain layer. **Consistent world-state updates** across independent nodes.
- **Docker-based adversarial simulation** with miners, clients, honest nodes, and malicious bots to validate the implementation against the attack scenarios studied.

Embedded distributed synthesizer — C, POSIX, Embedded, Real-Time Audio

- **Distributed audio system** composed of an Arduino controller, a Raspberry Pi audio engine, and an Android client for streaming and waveform visualization.
- **Low-resource modular real-time audio engine** in C using miniaudio, supporting polyphony, wave generation, modulation and filtering.
- **Controller-to-engine communication model** over serial, compacting physical inputs from the sensors into validated packets then polled and normalized into transformations to the generated wave.
- **Lock-free concurrency model** leveraging atomic global state variables and ring buffers for event and packet handling.
- **UDP audio streaming** to Android, using AudioTrack playback and an oscilloscope visualizer.

ACHIEVEMENTS & AWARDS

2nd Place – IEEE RetroJam Game Development Competition — IEEE UP Student Branch — Rust

2025

- Collaborated in a 3-person team to design and implement a complete **2D game from scratch** within **48-hour deadline** in Rust using only Raylib as support; personally composed the **original soundtrack**, designed in-game **audio effects**, and handled final **audio mixing and mastering**.

Academic Merit Awards — Escola Secundária Carlos Amarante

2020–2022

- Recognized for outstanding academic performance in science and technology disciplines throughout high school.

World 4th Place – RoboCup Rescue Superteam — RoboCup Leipzig, Germany

2016

- Represented Portugal internationally in **autonomous robotics** competition, developing **navigation algorithms** and **autonomous systems** for **rescue** scenarios.

National 1st Place – Robotics Championship (CoSpace Rescue) — Portugal

2016

National 4th Place – Canguru Matemático Sem Fronteiras — Portugal

2016

TECHNICAL SKILLS

Programming Languages: Rust, C/C++, Haskell, Python, Java, SQL

Frameworks & Tools: Git, Docker, UNIX, Tokio, POSIX, MPI, OpenMP, gRPC, LibP2P, Dalek, Raylib, PostgreSQL